



A systematic review of digital literacy in lifelong learning for older adults: challenges, strategies, and learning outcomes

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Abstract

Developing digital literacy (DL) is essential for older adults to keep pace with the rapid advances of technologies. This review took an evidence-based approach to examine the effectiveness of digital training programs designed for older adults. Adopting the PRISMA guideline, a systematic search was conducted on Web of Science, Scopus, and EbscoHost, which yielded a total of 4552 empirical articles. Of these, 46 were further examined based on the DigComp 2.2 Framework to identify digital competencies for older adults. Results showed that wealthier nations have greater technological adoption, with their older adults having better access to learning resources and training. Despite various barriers faced by some older adults, it would be beneficial to design exemplary solutions that group learners with similar levels of digital competency. This review identified that providing step-by-step instructions and pacing the lessons carefully can greatly enhance the learning experience. Other approaches, such as intergenerational learning, personal tutoring, game-based learning, dialogic learning, and peer learning, can effectively address the cognitive, social, attitudinal, and health-related needs of older adults. In future lesson design, it is worth implementing training on a long-term basis, personalizing the learning experience, and eliminating any practical barriers that might hinder the learning process. Moreover, future research should consider geographic diversity when recruiting participants, customize lessons for different cultural backgrounds, integrate technologies further into the daily life of older adults, and examine how emerging technologies could enhance the health of older adults.

Keywords Digital literacy · DigComp 2.2 Framework · Older adults · Gray digital divide · Systematic review

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Introduction

Digital technologies permeate almost every aspect of our lives. As digital technologies and our life become increasingly intertwined, developing digital literacy becomes essential for us to adapt to the digital world (Chu et al., 2021; Yu et al., 2017). Digital literacy (DL) is defined as the ability to use electronic devices and resources to investigate, organize, create, and communicate (Mohammadyari & Singh, 2015; Ng et al., 2023; Wang & Wu, 2022; Wedlake & Bugre, 2023). UNICEF (2019) identifies DL as a set of knowledge, skills, and attitudes that enable individuals to navigate the digital world effectively. It is relevant to learners of all ages, highlighting the need for comprehensive lifelong digital education (Meyers et al., 2013).

Meanwhile, the global population is aging at an accelerated pace. According to the World Health Organization, the proportion of people over the age of 60 is projected to nearly double by 2050, from 901 million (12%) to 2.1 billion (22%) (WHO, 2024). While 60% of EU residents over the age of 65 use the Internet and own mobile devices (European Commission, 2022), studies show older adults are less digitally literate than younger generations (Seifert et al., 2018; Tomczyk et al., 2022). With varying levels of digital competence, older adults belong to a growing trend known as “the gray (or elderly) digital divide”, which distinguishes the senior cohort of society who have knowledge and means to access information and communication technologies (ICT) and those who have not (Xu et al., 2024; Zhang et al., 2023). The COVID-19 pandemic further exacerbated the gray digital divide as stay-at-home orders and social distancing measures made the elderly homebound and led to their protracted isolation from family and friends (Finkelstein et al., 2023). This confirmed the findings in McMellon and Schiffman’s (2002) study, which linked social isolation among the elderly to the emergence of negative emotions like loneliness and depression, as well as mental health problems. As a result, digital literacy programmes for older adults have been designed to enhance the digital skills and strengthen the social connections of older adults, especially since the pandemic (McGinty, 2020).

However, at the same time, many older adults are grappling with a host of barriers that make it hard for them to fully leverage digital technologies in everyday life. On one hand, a decline in cognitive functions like memory and executive functions affects their mind from processing information. On the other hand, a gradual impairment of the senses like movement and balance makes it difficult for their body to operate an electronic device without physical assistance (Gitlow, 2014). Furthermore, personal circumstances such as financial stress and lack of family assistance prevent low-income brackets of older adults from owning electronic devices, eliminating their chances of access to the Internet and social media (Woodward et al., 2011). Even if local governments and non-profit organizations could provide health and financial support for older adults from lower socioeconomic backgrounds, they could meet other difficulties, especially psychological aspects. For example, a challenge faced by older adults is their wrestling of a range of emotions from fear, frustration and perception-induced pressure to curiosity and accomplishment, when trying to learn digital tools and platforms (Hage et al., 2020). These emotions, when wired with a lack of digital confidence, prior knowledge and skills, would make them a potential target for online scams (Nedeljko et al., 2022; Payne, 2020). Therefore, with meaningful pedagogical design and instructional support, older adults can become active agents to own their lifelong learning journeys.

Recent DL literature has provided many diversified learning strategies for older adults, such as age-appropriate digital content and applications, intergenerational pedagogies,

and user-centered design methods (e.g., Barrie et al., 2021; Bossio & McCosker, 2021; Oh et al., 2022; Tomczyk et al., 2022). However, reviews on the effectiveness of digital training delivery in developing and enhancing digital skills for older adults remain limited. For example, Oh et al. (2022) conducted a systematic review of DL measurements but the findings soon revealed a dearth of studies with suitable instruments. Schirmer et al. (2023) emphasized the need for learning programs to be tailored to older adults' specific challenges, identifying 47 publications that merely outline design criteria for promoting digital competence with no further evidence in support. Meanwhile, Nedeljko et al. (2022) identified three key categories: (1) poor DL among older adults, (2) inequality in ICT access, and (3) the role of ICT in mitigating social isolation among the 20 selected studies, but no feasible action plan was come up with to address the gaps identified therein.

To address the gaps, this article reviews the relevant DL literature so as to discern the essential learning components aimed at older adults, challenges they encountered, strategies employed to help them learn new skills, and impacts of those strategies on their technology attitudes, skill development, and technology usage habits. Five research questions are accordingly developed as below:

1. What are the characteristics of the research conducted on DL in lifelong learning for older adults?
2. What should older adults learn in DL learning programs?
3. What are the challenges faced by older adults when learning DL?
4. What strategies did prior research adopt in DL learning programs for older adults?
5. What are the major learning outcomes identified across these strategies for older adults?

Method

This article follows the Preferred Reporting Items for Systematic Evaluation and Meta-Analyses (PRISMA) guideline to review relevant publications (Tricco et al., 2018). The systematic review process can be conceptualized as: (1) identification, (2) screening, (3) eligibility, and (4) inclusion. The following will detail the systematic review procedures.

Identification

This first phase involves identifying articles for review. We conducted a systematic search in March 2024 across three well-known research databases: Web of Science, Scopus, and EbscoHost (Mongeon & Paul-Hus, 2016). The search terms were identified based on prior reviews on older adults' DL (Ahmad et al., 2022; Farias-Gaytan et al., 2022). The search strategy utilized Boolean logic to scan titles, abstracts, and keywords for publications between January 2014 and January 2024 (Table 1). Given that older adults' DL in lifelong learning is multidisciplinary, the relevant research areas were then selected in the databases accordingly (see Table 2). This study defines older adults as individuals aged 60 and above, in accordance with the definition set out by the World Health Organization (WHO, 2024).

Table 1 Search terms used in database searches

Topic	Search terms
Older adult	("old* adult*" or senior* or elderly or "young old")
Digital literacy	("digital literacy" or "information and communication technology" or "ICT" or "information technology" or "social media" or Internet or computer* or "digital competence*" or "digital skill*" or e-skill* or "Digital fluency" or "digital technology*" or digital* or gadget* or techno*)
Education	(educat* or learn* or teach* or course* or program* or intervention*)

Note To ensure the search results contain the key terms, the authors used Boolean operators: "OR" to unite multiple terms and "AND" to join them. The specific search terms were adapted to the syntax of each database

Table 2 Areas selected in each database

Database	Areas
Web of Science	Social sciences interdisciplinary, education educational research, gerontology, geriatrics gerontology, education scientific discipline, linguistics, and communication
Scopus	Social sciences
EbscoHost	Aging & older adult development, curriculum & programs & teaching methods, educational & school psychology, specialized interventions, professional education & training, classroom dynamics & student adjustment & attitudes

Screening, eligibility, and inclusion

After completing the search, screening was conducted to determine whether articles contained material that would be relevant for this systematic review. A series of inclusion and exclusion criteria were set to screen articles (see Table 3). Articles should be related to the effects of learning processes on older people supported by DL and were published in the period between April 2014 and April 2024. Inspired by the local situation in Hong Kong, where the proportion of elderly adults aged 65 and above using the internet has increased more significantly by 24.0% in 2014 than before (Hong Kong Census & Statistics Department, 2020), we aim to explore DL in lifelong learning among older adults and summarize the publication trends over the last ten years. Furthermore, the articles should be reported in English, featuring a learning technology intervention and documented as empirical research. Additionally, the full-text of articles should be made available, and the papers should be published in peer-reviewed journals. If there was disagreement as to whether an article should be excluded, it would be resolved through discussion with the co-authors to make a decision. In this phase, articles that were not relevant to the keywords/phrases of "older adults", "digital literacy", and "education for older adults" were excluded.

Table 3 Inclusion and exclusion criteria

	Inclusion criteria	Exclusion criteria
Theme	The article was related to the effects of a DL learning program on older adults' attitudes, DL development, and technology usage habits	The article was not concerned with the effects of a DL learning program on older adults' attitudes, DL development, and technology usage habits
Publication period	The publication date was between April 2014 and April 2024	The publication date was earlier than April 2014 or later than April 2024
Language	The reporting language was in English	The reporting language was not in English
Intervention	The research involved a learning intervention that incorporated digital technologies or tools to facilitate digital literacy	The research did not involve a learning intervention that incorporated digital technologies or tools to facilitate digital literacy
Type of study	The research was an empirical study	The research was intended as a discussion paper or a review article
Availability	Full-text was available	Full-text was not available
Peer review	The article had gone through a peer review process	The article had not gone through a peer review process

Following the initial screening, the eligibility of the studies was assessed to determine whether the remaining articles would help answer the research questions. This assessment was reviewed based on the full texts of the articles. The first and second-named authors read the full text of an article and made an “include” or “exclude” decision on each article. Based on the three database searches, the initial database search using the terms described in Fig. 1 identified 8672 articles. After removing 4120 duplicates and irrelevant articles, 4552 remained. Further screening excluded 1282 articles not related to “older adults”, 1472 not related to “digital literacy”, and 1674 not related to “older adults’ learning”, leaving 124 articles. An additional 78 articles were removed for not reporting empirical research or not matching the key criteria. Ultimately, 46 articles were selected for further thematic analysis.

Data extraction and analysis

After exporting the 46 selected articles, a thematic analysis was conducted in accordance with the guideline outlined by Clarke and Braun (2017). We first familiarized ourselves

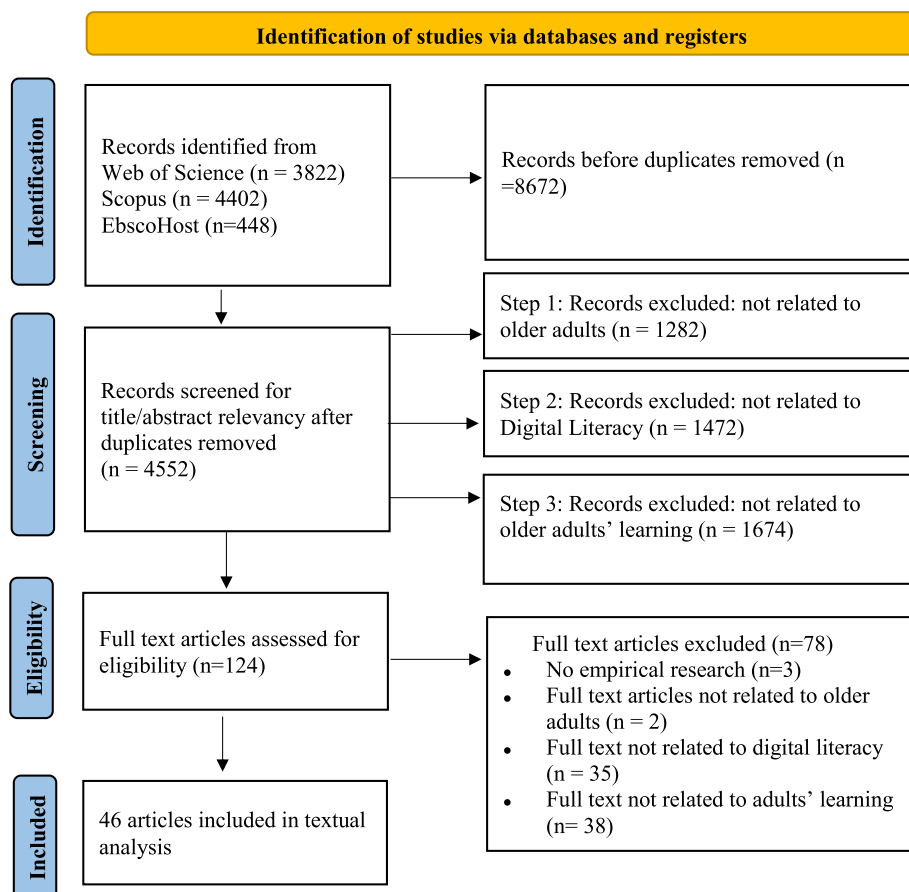


Fig. 1 PRISMA flowchart for the study identification process

with the articles and had a quick read through each of them. We then generated initial codes, systematically coded the data, and grouped them into themes and took note of meaningful patterns. Afterwards, we refined the themes as needed and assigned a name to each theme in a discrete manner. Finally, we provided a detailed analysis of the themes and texts, drawing relevant textual evidence from the articles. This iterative process enabled us to identify useful text segments and organize these segments into different patterns to address the study's RQs.

To address RQ1, we extracted data from each article on the author(s), year of publication, countries/regions in which the intervention took place, and research methods (i.e., quantitative, qualitative, and mixed methods). To address RQ2, learning components were extracted from DL programs, focusing on the five key areas outlined in the "DigComp 2.2: The Digital Competence Framework for Citizens" ("the DigComp 2.2 Framework"), namely (a) information and data literacy, (b) communication and collaboration, (c) digital content creation, (d) safety, and (e) problem-solving. This model serves as a comprehensive guide to equip citizens, including older adults, with the necessary digital skills and knowledge to navigate the digital world. To address RQ3 to RQ5, the findings/results and discussions of the selected studies were analyzed thematically using a deductive thematic approach to study the difficulties faced by older adults in the learning programs, approaches mentioned in these studies, and learning impacts of these programs on older adults.

To ensure coding reliability, the first author and an external researcher were involved in the coding step, and each article was independently reviewed, with a 92% agreement rate exceeding the suggested 80% consistency threshold for key findings in each article under review (Miles & Crisp, 2014). After several rounds of discussion and further scrutiny of the studies, we reached a consensus over the reliability of the codes and presented our results in the subsequent sections.

Results

Characteristics of research on DL in lifelong learning for older adults (RQ1)

Year of publication Figure 2 summarizes the publication trends of selected studies from 2014 to 2024. As highlighted in the figure, the average number of paper outputs rose from around 2.0 in 2014–2016 to around 5.3 in 2022–2024, revealing an upward trajectory during this period. In recent years, notable increases in publications can be observed when compared with previous years (e.g., 2014–2016 and 2018). One possible reason is that people become more aware and proactive in promoting digital inclusion measures aimed at empowering older adults to master digital technologies. It can be attributed to the COVID-19 pandemic according to Finkelstein et al.'s study (2023), which brought the issue of gray digital divide in many societies to the fore and drove many older adults to connect with others through online social media and rely more on online services for their daily living (e.g. for shopping and seeking healthcare information). Additionally, the launch of OpenAI's generative artificial intelligence (GenAI) chatbot ChatGPT in November 2022 has spurred interest on the effects of GenAI technologies on older adults, particularly DL development (Kamoun et al., 2024; Sriwisathiyakun & Dhamanitayakul, 2022). One example underscoring the exigency of DL development for older adults is the frequency at which older adults are victimized in common cyber scams due to lack of digital awareness, skills and knowledge, highlighting the necessity of educational initiatives aimed at enhancing the relevant

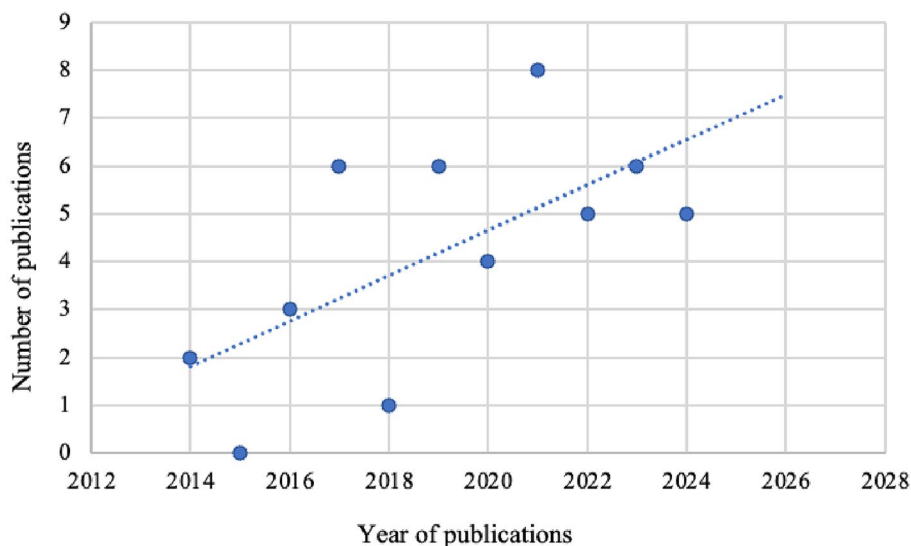


Fig. 2 Publication trends of selected studies from 2014 to 2024 (n=46)

DL areas. This combination of factors has led to a greater focus on understanding how older adults interact with and adapt to these rapidly evolving technologies.

Geographic diversity Figure 3 shows the countries and regions where the intervention took place. A total of 46 studies* originated from five continents: North America (n=24,

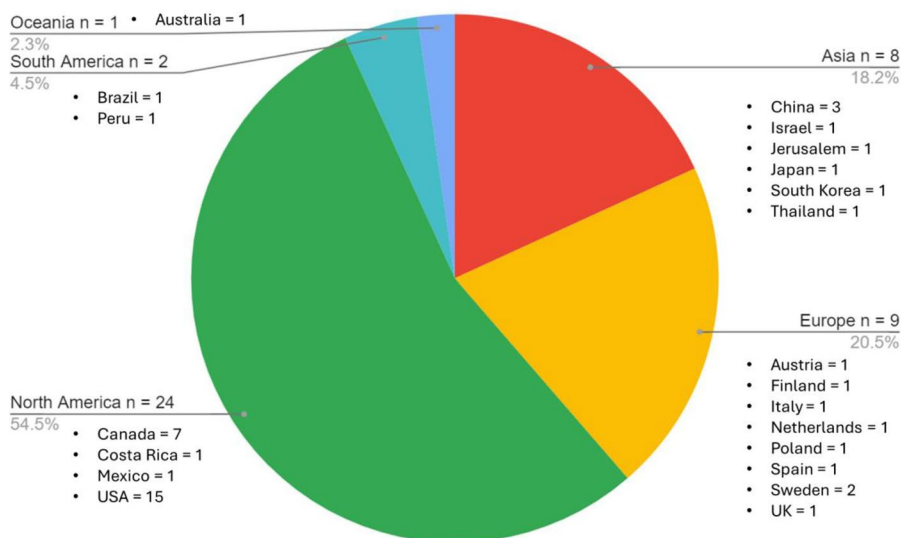


Fig. 3 Countries and regions where interventions were implemented (n=46). *Note* Two studies were excluded from the list as the interventions adopted in the two studies took place across multiple countries/regions

54.5%), Europe (n=9, 20.5%), Asia (n=8, 18.2%), South America (n=2, 4.5%), and Oceania (n=1, 2.3%). There were notable differences in the regional distribution of these studies. Significantly, no studies came from Africa, and only a few originated from South America and Oceania.

Research methods Table 4 summarizes the research methods used to explore digital education for older adults. Of the 46 articles examined, 18 (41.3%) employed quantitative methods, using data collection procedures like surveys and questionnaires to examine learning aspects such as habits, motivation and engagement (e.g., Chan et al., 2016; Myhre et al., 2017; Vaportzis et al., 2017). 11 studies (23.9%) used qualitative methods, including interviews and observations, to understand older adults' learning experiences and challenges (e.g., Baric et al., 2019; Ferreira et al., 2016; Hage et al., 2020). The remaining 17 papers (34.8%) took a mixed-methods approach, combining qualitative and quantitative techniques to provide a more comprehensive perspective (e.g., Zhong et al., 2022).

Key competencies that older adults learn in digital learning programs (RQ2)

Based on the DigComp 2.2 Framework (2022), this section summarizes five key competencies of digital learning programs for elderly citizens as highlighted in Table 5. Among the five competencies reviewed, **information and digital literacy** was the most mentioned DL skill for older adults (n=84). It measured how older adults effectively access, evaluate, and manage information online (Miwa et al., 2017). The skills necessary to meet this competency encompass a range of abilities, from using search engines and web browsers to navigating online platforms (Myhre et al., 2017), and performing everyday tasks on mobile devices, such as taking photos (Zhang et al., 2023) and checking emails (Kim et al., 2022). Most of these articles focused on teaching older adults how to effectively use their mobile/computer devices (n=38) and mobile applications for their everyday tasks (n=34).

The second most commonly mentioned DL skill was **communication and collaboration** (n=34), which requires elderly citizens to use various communication tools to connect with family members, friends and others (e.g., social workers and adults with disabilities) (Ferreira et al., 2016; Tsugawa, 2023). The third mentioned skill across DL interventions was **problem solving** (n=24), which requires older adults to learn the basic functions of a digital device (Saravanan et al., 2024) and access online public services such as settling utility bills or using e-banking services (Korpela et al., 2023; Liang et al., 2023). Additionally, seven studies (n=7) focused on troubleshooting basic computer-related issues, helping older adults address tablet usage and network problems. Another seven studies (n=7) examined the application of digital skills to solve everyday challenges, including learning to use various mobile applications effectively. The last two mentioned topics were **digital safety** (n=10) and **digital content creation** (n=4), which are respectively concerned with how older adults protect their privacy (Ferreira et al., 2016; Saravanan et al., 2024; Seo et al., 2019) and how they create digital creative contents through photo- or video-editing tools and online music reels (Arthanat, 2021; Delello & McWhorter, 2017; Ferreira et al., 2016).

The digital literacy studies for older adults reveal a comprehensive array of 22 different types of technologies that play a crucial role in enhancing their key competencies. These technologies encompass a variety of devices and applications, such as tablets and mobile apps (n=7), which facilitate everyday tasks and problem-solving. The Internet serves as a foundational resource for accessing information, while email and social media platforms (n=4) enable effective communication and connection with others. Additionally, tools for

Table 4 Research methods used in digital literacy in lifelong learning for older adults

Research methods	Representative examples	Num- ber of articles	Sample studies
Quantitative	In the study conducted by Kim et al. (2022), researchers investigated the impact of an intervention aimed at promoting Internet use on communication patterns, online activity, and social connections in elderly individuals with limited financial resources. The research included a total of 77 participants who underwent a 12-week training program focusing on computer knowledge as well as Internet knowledge. The results indicated that the advantages of the intervention were particularly notable for marginalized populations	18	Chan et al. (2016), Myhre et al. (2017), Vaportzis et al. (2017)
Qualitative	Zhang et al. (2023) investigated the correlation between social networking and the utilization of information and communication technology among older people. Data were gathered through interviews and participant observation conducted during 29 sessions. The finding of the study highlighted the crucial role of technological support in the development and enhancement of the participants' social networks	11	Baric et al. (2019), Ferreira et al. (2016), Hage et al. (2020)
Mixed methods	Blažič and Blažič (2020) analyzed the efficacy of contemporary methodologies in enhancing the digital literacy of elderly individuals, shedding light on the obstacles and perceptions they encounter. The research involved 60 older adults who participated in gaming activities or training sessions, followed by interviews and questionnaire completion. The researchers emphasized the necessity of enhancing educational programs to address the digital divide prevalent among this demographic group	17	Barbosa Neves et al. (2019), Delello and McWhorter (2017), Leek (2021)
Total	—	46	

Table 5 Five key competencies identified in digital learning programs based on the DigComp 2.2 Framework (European Commission, 2022)

Key competencies	Definitions (European Commission, 2022)	Digital technologies adopted in the selected studies
Problem solving (n = 24)	For the purpose of recognizing requirements and issues and tackling theoretical matters and obstacles in digital settings. To employ digital instruments to revolutionize procedures and goods and to remain updated on digital progressions	• Troubleshoot basic computer related issues e.g., tablets usage and network problems (n = 7) • Apply digital skills to solve everyday problems (e.g., learning how to manipulate with various mobile apps) (n = 7) • Learn to use online tutorials and support resources e.g., navigating the Internet and accessing online resource (n = 6) • Adapt to digital tools in everyday lives (e.g., email and social media) (n = 4)
Digital safety (n = 10)	To safeguard devices, content, personal data, and privacy in digital environments. To protect physical and mental health while recognizing the role of digital technologies in promoting social well-being and inclusion. To understand the environmental impact of digital technologies and their usage	• Identify and avoid online scams, phishing attempts, and malware (n = 4) • Practice safe online behaviors, such as online banking and shopping (n = 3) • Understand the importance of personal data privacy and protection (n = 2) • Secure devices with strong passwords and update software regularly (n = 1)
Digital content creation (n = 4)	To create and edit digital content. To enhance and integrate information and content into an existing body of knowledge while understanding how to apply copyright and licenses. To provide clear instructions for using a computer system	• Capture and edit digital photos, audio, and video (n = 2) • Explore digital creativity tools and applications (n = 1) • Create and edit basic text documents, spreadsheets, and presentations (n = 1)
Information and digital literacy (n = 84)	To express information needs, locate and retrieve digital data, information, and content. To assess the relevance of sources and their content. To store, manage, and organize digital data, information, and content	• Effective use of mobile/computer devices e.g., Ipad, computer (n = 38) • Effective use of mobile applications e.g., Zoom, Facebook, Google Street View, email and photo sharing (n = 34) • Effectively search for and identify reliable online sources of information (n = 5) • Entertainment themselves (e.g., playing computer games) (n = 5) • Organize and manage digital files and documents (n = 1) • Interpret data visualizations and infographics (n = 1)

Table 5 (continued)

Key competencies	Definitions (European Commission, 2022)	Digital technologies adopted in the selected studies
Communication and collaboration (n = 34)	To engage, communicate, and collaborate using digital technologies while recognizing cultural and generational diversity. To participate in society through both public and private digital services and active citizenship. To manage one's digital presence, identity, and reputation	• Use communication tools like email, social media, video/audio calling and instant messaging software to stay connected and collaborate (n = 15) • Facilitate meaningful interactions and communication between older adults and others by using social media (e.g., individuals with intellectual disabilities, social workers, family members and peers) (n = 15) • Foster effective communication and collaboration between older adults and children by using social media (n = 3) • Communicate with others via web-conferencing software (e.g., Zoom) (n = 1)

Notes 1. The frequency reflects cumulative coverage that exceeds the number of selected papers, as some studies address multiple competencies. 2. *n* = frequency; the sum of the frequencies for this question exceeds the number of papers selected because some studies refer to more than one competency

online banking promote financial literacy and security in digital transactions. The inclusion of multimedia technologies, such as digital photos, audio, and video, highlights the creative potential these tools offer ($n=3$). Furthermore, participants engage with essential software for document creation, including text documents, spreadsheets, and presentations, alongside popular applications like Zoom and Facebook. Other technologies, such as Google Map and computer games, provide opportunities for exploration and entertainment.

Challenges faced by older adults when learning DL (RQ3)

Table 6 presents five major challenges faced by older adults when learning DL. Among the selected studies, 60.87% ($n=28$) highlighted a lack of basic **digital knowledge and skills** among older adults (Leek, 2021; Saravanan et al., 2024; Stanley et al., 2019). Additionally, 45.65% ($n=21$) identified **physical and cognitive decline** as significant hurdles (Castro Rojas, 2021; Gamliel & Gabay, 2014; Leone et al., 2018; Tellado et al., 2024). **Emotional factors**, including fear, anxiety, frustration, and reduced self-efficacy, were mentioned in 36.96% of studies ($n=17$) (Arthanat, 2021; Hage et al., 2020; Neil-Sztramko et al., 2020; Tsugawa, 2023). Furthermore, 19.57% of studies ($n=9$) noted a **lack of access to new technologies and unsupportive environments** as barriers to improving DL among older adults. Addressing these challenges is essential for enhancing digital skills and adoption in this demographic (Baker et al., 2017; Chen et al., 2021; Kim et al., 2022). Lastly, 17.4% of studies ($n=8$) referenced negative **social influences**, such as unsupportive peers and social prejudice, which impede learning motivation (Seaton et al., 2023; Seo et al., 2019; Wang et al., 2024).

Strategies that prior research adopted in DL learning programs (RQ4)

Table 7 summarizes the learning strategies adopted by older adults in various DL learning programs. Of the 46 articles scrutinized in this study, 41.3% of studies ($n=19$) presented **lecture-based instruction** as an approach to learning DL. Lecture-based instruction is a traditional mode of teaching where the instructor teaches course contents to students in a classroom setting through visual aids like PowerPoint slides and handouts (Kantar & Sailian, 2018). Secondly, **intergenerational learning and personal tutoring** were each utilized in 15.22% of studies ($n=7$). Intergenerational learning involves an exchange of knowledge, innovation, and experience between at least two age cohorts, such as teenagers/children and older adults (Newman & Hatton-Yeo, 2008). Personal tutoring refers to one-on-one learning sessions between older adults and a teacher to support their individual learning needs (Chen et al., 2021). Thirdly, 8.70% of studies ($n=4$) implemented **game-based learning**, which achieves learning outcomes through engaging game content and gameplay, facilitating problem-solving, and providing learners with a sense of accomplishment (Qian & Clark, 2016). Fourthly, 4.35% of studies ($n=2$) adopted a **dialogic learning approach**, which leverages dialogues to promote learning and cognitive development, emphasizing collective problem-solving and the integration of diverse student voices, thinking and knowledge (Alexander, 2008). Finally, 2.17% of studies ($n=1$) utilized **peer learning**, a strategy that encourages learners to support one another in their learning journeys (Boud & Cohen, 2014).

Table 6 Summary of challenges faced by older adults when learning DL (Kebede et al., 2022)

Challenges	Descriptions	Example studies
Technical knowledge and skills (n = 28)	Lack of knowledge and skills to use digital technology effectively	Saravanan et al. (2024), Stanley et al. (2019)
Physical and cognitive limitations (n = 21)	Physical limitations and cognitive changes associated with aging	Castro Rojas (2021), Gamliel and Gabay (2014), Leone et al. (2018), Tellado et al. (2024)
Emotional factors (n = 17)	Feelings (e.g., anxiety) towards digital technology	Arthanat (2021), Hage et al. (2020), Neil-Sztramko et al. (2020), Tsugawa (2023)
Environmental context and resources (n = 9)	Lack of access to digital technologies due to factors such as poverty and living in areas with poor network connectivity	Baker et al. (2017), Chen et al. (2021), Kim et al. (2022)
Social influences (n = 8)	Lack of support from family, peers, and society	Seaton et al. (2023), Seo et al. (2019), Wang et al. (2024)

Note The total frequencies exceed the number of papers because some studies addressed multiple challenges

Table 7 Strategies adopted in digital learning programs for older adults

Strategies	Descriptions	Num- ber of articles	Example studies
Lecture-based instruction	Traditional teacher-student training provided informative and age-conscious lessons on digital skills, aiming to bring about older adults' confidence in using technologies	19	Miwa et al. (2017), Ferreira et al. (2016), Gatti et al. (2017), Tsai et al. (2019)
Intergenerational learning	Intergenerational programs fostered mutual understanding and respect between older adults and younger generations, enhancing the learning experience through personal sharing	7	Aldaco-Arias and Silas-Casillas (2021), Gamliel and Gabay (2014), Leek (2021)
Personal tutoring	Tutoring programs facilitated the uptake of digital skills that reduced social isolation among older adults	7	Chen et al. (2021), Ferguson et al. (2024), Fields et al. (2021)
Game-based learning	Game-based programs stimulated older adults with an engaging and interactive user experience	4	Blažič and Blažič (2020), Jurczyk-Romanowska et al. (2014)
Dialogic learning approach	Programs, which promoted bilateral dialogues between older adults and others, reduced digital divides	2	Taieb-Maimon and Vaisman-Fairstein (2022), Tellado et al. (2024)
Peer learning	Programs drawing on instant messaging technologies leveraged the power of social networks to enhance peer-to-peer learning	1	Korpela et al. (2023)

Learning outcomes of DL learning programs for older learners (RQ5)

Table 8 summarizes the learning outcomes identified in the selected DL programs for older learners. It highlights 4 categories: cognitive, social, attitudinal/emotional, and health-related. Of the studies reviewed, the most frequently measured learning outcome was **cognitive**, which refers to the development of digital knowledge and skills, alongside information literacy skills, with 64 articles focusing on the theme of **DL**, which encompass competencies, skills, cognitive abilities, and brain activity ($n=38$), and **information literacy**, which involves evaluating and using information effectively ($n=26$). The second most measured outcome was **social**, which refers to the enhanced social engagement and relationships with others and accounts for 38 articles dealing with **social engagement** ($n=30$) and **strengthening relationships with family and friends** ($n=8$). The third outcome was **attitudinal/emotional**, which refers to the improved attitudes and emotional responses toward DL learning and has a total of 36 articles reporting on **positive attitude toward ICT** ($n=13$), **confidence and self-efficacy** ($n=13$), **motivation** ($n=6$), and **overcoming fear of learning technologies and anxiety** ($n=4$). The last outcome was **health-related**, which deals with improvement to one's mind resulting from participating in DL activities and which is concerned primarily with **mental health** ($n=11$) and **cognitive health** ($n=7$).

Summary of findings

This review investigates digital literacy related to the elderly's lifelong learning through five research questions (RQs). RQ1 summarizes the background information of current DL studies about lifelong learning for older adults in terms of the publication years, geographic diversity, and research methodologies of the DL literature. RQ2 suggests the use of DigComp 2.2 Framework (2022) to identify key competencies essential for DL programs targeting older adults. Five common digital tasks that older adults should learn are found: problem solving, digital safety, digital content creation, information and digital literacy, and communication and collaboration. RQ3 highlights the challenges older learners face in acquiring digital literacy. These challenges encompass a range of factors, including technical knowledge and skills, physical and cognitive limitations, emotional barriers, environmental contexts and resources, and social influences. These are useful for instructional designers and researchers to design appropriate learning strategies which are outlined in RQ4, which include lecture-based instruction, intergenerational learning, personal tutoring, game-based learning, a dialogic learning approach, and peer learning. RQ5 summarizes what older adults can learn in four dimensions: cognitive, social, attitudinal/emotional, and health-related. To gather the results identified in the five RQs, Figure 4 illustrates a framework to summarize the findings.

Discussion

The characteristics of the DL in lifelong learning research for older adults

Year of publication Regarding the year of publication, there are almost three times as many selected studies in 2024 as compared to 2014. In 2024, the number of smartphone users worldwide reached 4.3 billion, representing over half (54%) of the global population.

Table 8 Summary of effectiveness of the DL learning programs on older learners

Measured outcomes	Themes	Frequency	Descriptions	Example studies
Cognitive (n = 64)	Digital literacy	38	Older adults possess digital knowledge and skills needed to operate a digital device securely, efficiently, and with confidence	Pachis and Zonneveld (2019), Seo et al. (2019), Taha et al. (2016)
	Information literacy	26	Older adults possess the ability to recognize when to find information and how to locate, evaluate, and use the needed information effectively	Moore and Hancock (2022), Gatti et al. (2017), Myhre et al. (2017), Seo et al. (2019)
Social (n = 38)	Social engagement	30	Older adults leverage digital technologies to socialize, collaborate, and communicate with the wider community	Delello and McWhorter (2017), Gatti et al. (2017), Myhre et al. (2017)
	Promotion of family and friends' relations	8	Older adults stay connected with family and close friends via social apps	Neil-Sztramko et al. (2020), Stanley et al. (2019)
Attitudinal/emotional (n = 36)	Attitude	13	Older adults feel positive towards ICT	Saravanan et al. (2024), Stanley et al. (2019)
	Confident and self-efficacy	13	Older adults feel informed and self-efficient when using technologies	Gamliel and Gabay (2014), Wang et al. (2024)
Health-related (n = 18)	Motivation	6	Older adults are motivated to use digital technologies	Gatti et al. (2017), Tsugawa (2023)
	Overcoming fear and anxiety towards digital learning	4	Older adults overcome fear and anxiety towards digital learning	Chen et al. (2021), Ferreira et al. (2016)
	Mental health	11	Digital technologies help older adults reduce social isolation, anxiety, and boost their happiness	Chen et al. (2021), Fields et al. (2021), Liang et al. (2023)
	Cognitive health	7	Digital technologies enhance older adults' cognitive abilities (e.g. executive function, processing speed and episodic memory)	Castro Rojas (2021), Ten Brinke et al. (2020)

Note The frequency reflects the cumulative coverage that exceeds the number of selected papers as some studies address multiple learning outcomes

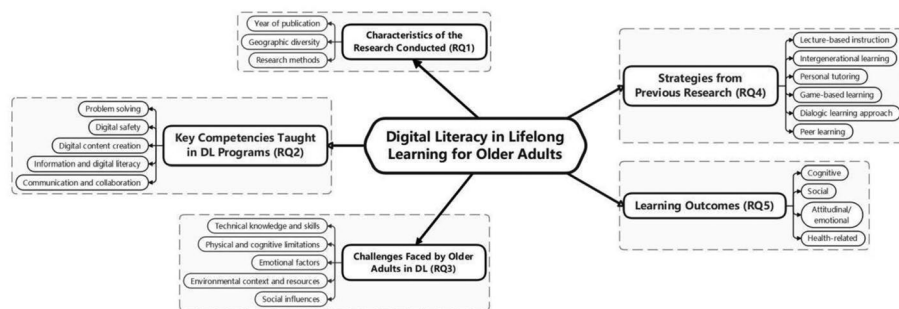


Fig. 4 A framework to summarize the findings

This marks a significant increase from 2014, when smartphone penetration only numbered at 1.76 billion, accounting for 25% of the global population (eMarketer, 2014; GSMA Report, 2023). As the global aging issue intensifies, there is growing demand for older adults to learn digital knowledge to enhance their well-being and living standards, and to address everyday challenges that now come digitally through technologies. Coupled with the COVID-19 pandemic, which has accelerated the consumers' shift to online services and communications, it has highlighted the essential need for DL in lifelong learning among older adults. Moreover, new cyber threats coming from AI and AI-generated deep-fake contents pose new challenges for older adults and require up-to-date DL in lifelong learning to keep them alert to dubious online information (Brashier & Schacter, 2020). This called for the need for all citizens including older adults to equip themselves with a certain level of AI literacy (Ng et al., 2021, 2024).

Geographic diversity With the vast amount of resources and financial support for citizens living in the upper-middle-income countries (39 out of 46 studies), the geographic distribution analysis reveals important gaps in research from developing regions. Addressing these gaps is crucial for ensuring that DL efforts are inclusive to highlight the need for targeted interventions, and reduce barriers to digital access and education. Understanding the unique cultural and economic contexts of these areas can inform the design of effective DL programs tailored to their needs. Moreover, a great majority of older adults in these areas have achieved a level of DL that is often unmatched by older adults in lower-income countries (Mhlana, 2024). This aligns with most findings of this review that were designed and implemented in North America, which account for 50% of the total reviewed studies. It also aligns with the current global statistics of digital populations. According to a national survey, of the approximately 7.21 billion active smartphones around the world, nearly nine out of ten adults in the U.S. (89% of those aged 50 and older and 90% of those aged 18–49) owned a digital device (Kakulla, 2024). In contrast, only 18.4% to 46.5% of the population in certain low- to middle-income countries, such as India, Indonesia, and Pakistan, own a digital device (World Atlas, 2020). With limited studies completed in Africa, Oceania, and South America, as illustrated in section “[Characteristics of research on DL in lifelong learning for older adults \(RQ1\)](#)”, it reflects a small number of digitally-literate elderly populations living in those regions and implies a high level of digital exclusion and alienation for most of them. Against such a societal backdrop, more DL research and resources are called for these communities.

Research methods Across the 46 reviewed studies, both quantitative and qualitative approaches were employed to capture measurable learning outcomes and the contextual complexities that shape the learning experiences of older adults. Quantitative methods utilized established questionnaires, such as the Duke Social Support Index, De Jong Gierveld Loneliness Scale, 12-item Lubben Social Network Scale, and Older Adults' Computer Technology Attitudes Scale, to help service providers evaluate their programs and assess the effects of their interventions. Qualitative methods, including interviews, content analysis, and thematic analysis, were used to explore participants' perceptions of learning interventions. By triangulating quantitative and qualitative findings, mixed methods provide valuable insights for program design and refinement, as well as for resource planning and allocation for older adults, such that no elderly person is left behind in the increasingly digitized society.

Proposed frameworks of DL for older adults

Five areas of digital literacy for older adults

As section “[Key competencies that older adults learn in digital learning programs \(RQ2\)](#)” reveals, few studies have established conceptual models to guide what older adults should learn. Zhang et al. (2023) presented a preliminary model identifying five major types of digital tasks older adults should learn: (1) communication, (2) entertainment, (3) everyday tasks, (4) public service, and (5) e-commerce. RQ2 aligns with this and goes beyond just identifying types of digital tasks. It also considers the ethical concerns older adults may face, addressing issues around online privacy, data protection, digital threats, misinformation, and critical evaluation of online information. The components identified in this review should be viewed through the lens of the DigComp 2.2 Framework, as older adults are in fact part of the broader digital citizenship. This study reframes the digital skills and competencies specially designed for older adults, with the aim of catering to their needs, expectations, and physical and cognitive limitations. Adopting this perspective underscores the significance of ensuring elder citizens are also part of the digital learning design, rather than continuing as a digitally-excluded group in society. It is hoped that older adults are empowered to take on a proactive role in their digital learning process by having access to the same set of learning tools and resources as the digitally-cognizant.

For the skills identified in RQ2, the findings are consistent with the recommendations made in other recent reviews in this area. For example, Myhre et al. (2017) found that DL instruments used in older adult studies can be examined through the lens of the European Commission's DigComp 2.2 Framework, addressing the elements of age-appropriate DL. Also, Kang et al. (2023) stressed the importance of “information and data literacy” and “communication and collaboration” for older adults in Korean DL studies, aligning with the current research focus on both operational skills and critical thinking using the DigComp model. On one hand, Arthanat (2021) showed that older people's participation in ICT was facilitated in their study, with specific activities including managing health through ICT, finding entertainment, and engaging with everyday activities. On the other hand, Neil-Sztramko et al. (2020) suggest that older people's participation in ICT programs encouraged themselves to find health information through tablets.

Furthermore, merely knowing how to use digital technologies for daily tasks may not be sufficient per se. Older adults tend to be vulnerable to cyber crime as previously discussed in section “[Key competencies that older adults learn in digital learning programs \(RQ2\)](#)”

about digital safety and section “[The characteristics of the DL in lifelong learning research for older adults](#)” about the trend of deepfake contents. As such, older adults must develop cybercrime awareness and a safe browsing mindset that enable them to distinguish between false/fake information and authentic contents, as well as to avert potential encounters of digital exploitation (Karagiannopoulos et al., 2021). Implications for becoming victims of digital frauds can be manifold, not the least of which is the adverse influence on older adults’ mental health (Burnes et al., 2017). Some reports have also highlighted the heightened level of negative emotions, such as depression, anxiety, and self-blame, among the elderly victims following incidents of digital crimes (Kircanski et al., 2018; Shang et al., 2022). Hence, equipping older adults with the skills to critically evaluate online content, detect scams, and make informed decisions is essential for the prevention of digital fraud. Fostering these higher-order digital competencies, beyond mere functional dimension, is an important aspect of technological upskilling for the elderly.

Digital technologies adopted in each area

We identified essential digital technologies for older adults across various areas of DL. Firstly, in information and digital literacy, classes often cover basic skills like turning devices on and off. While these topics may seem mundane, they are crucial for those unfamiliar with digital concepts. Previous research, such as Choudrie et al. (2020), highlights the essence of older adults interacting with smartphones, yet many lack skills in integrating these devices into daily tasks. Smartphones enhance communication and connectivity, emphasizing the need for tailored know-how courses for seniors to connect with their friends and family members. Older adults benefit from using social media, email, and platforms like Zoom to connect with family, peers, and caregivers. Research by Dennis et al. (2016) shows that maintaining contact with loved ones can significantly reduce loneliness, while Choudrie et al. (2020) note the joy older adults find in sharing experiences captured on smartphones.

Thirdly, problem-solving skills are essential for navigating technology. Older adults learn to tackle basic computer issues and apply digital skills to everyday tasks. For instance, Korpela et al. (2023) demonstrated how older adults use online banking and e-government services to manage finances and use public facilities. Taieb-Maimon and Vaisman-Fairstein (2022) enhanced older adults’ internet skills through real-time guides that provide feedback on their actions, fostering confidence in their digital skills. Moreover, digital security is crucial for developing DL. Older adults are particularly vulnerable to online threats, as highlighted by Moore and Hancock (2022), who found that they often struggle with misinformation and frauds. Teaching strategies to identify online scams is essential to avoid deceptions. Finally, while few studies address digital content creation, activities like creating and editing digital media foster cognitive engagement and creative expression. This is consistent with a study conducted by Choudrie et al. (2020) who found that mobile phones enable older adults to capture memories, enhancing their ability to recall past experiences.

Individual and structural factors in DL development

RQ3 highlights the various challenges older adults face in using digital technologies, emphasizing the need for effective strategies to empower this population to become digitally literate. The key barriers identified in section “[Challenges faced by older adults when learning DL \(RQ3\)](#)” can be synthesized into two main factors: **individual factors** (e.g.

personal technical skill gaps, physical and cognitive declines, and complex emotions) and **structural factors** (e.g. social influences and environmental resource constraints).

Individual factors

The rapid advancement of digital technology could create a significant digital divide for older adults. Saravanan et al. (2024) found that many older adults had never used tablets or computers before participating in their study, leading to difficulties in learning how to operate digital devices. Even those with some technical experience face challenges due to a diversity of digital technologies available on the online marketplace (e.g. tablets, smartphones, smartwatches, etc.), each with different brands, models, operating systems and user interfaces. These findings highlight a deficiency in digital skills and knowledge among older adults. Considering that they start on an unlevel playing field than the other age groups (e.g., see section “[Effective approaches to foster older adults’ DL](#)” on Intergenerational Learning) and cultural groups (e.g., see section “[Characteristics of research on DL in lifelong learning for older adults \(RQ1\)](#)” on Geographic Diversity), it is inevitable that they often require more time to master the more fundamental skills, are relatively more prone to making errors during the learning process, and generally require more hands-on assistance (Taieb-Maimon & Vaisman-Fairstein, 2022). To accommodate the learning pace and cognitive abilities of older adults, sufficient foundational instruction is essential. For example, intergenerational learning and one-on-one private training can provide more tailored feedback and support than the conventional lecture-based learning (Betts et al., 2019; Saravanan et al., 2024).

The challenges faced by older adults due to physical and cognitive declines are equally noteworthy. In the present study, twenty-one articles referenced “physical and cognitive impairments”, highlighting declines in hearing, visual acuity, motor skills, memory, and cognitive abilities that can negatively impact older adults from effectively learning digital skills (Baric et al., 2019; Hawthorn, 2000). For instance, Gamliel and Gabay (2014) observed that older adults often ask the same questions repeatedly when learning to use computers in intergenerational settings, reflecting their declining memory. This necessitates the repeated instructions from younger participants, entailing as much patience in the process as the adjustment of the pace and pronunciation of their speech. Similarly, Jurczyk-Romanowska et al. (2014) found that older adults may struggle to complete learning tasks due to hearing limitations (even with hearing aids), reading difficulties, and insufficient computer skills. However, Chan et al. (2016) demonstrated that participation in an iPad intervention program improved two cognitive functions: processing speed and situational memory. This suggests that engaging in such activities can support cognitive functioning and may help slow age-related cognitive decline.

A last challenge for older adults is their experience of complex emotions during lessons, as highlighted in 17 studies. Specifically, Chen et al. (2021) found that older adults often experience anxiety and nervousness when using tablets, primarily due to concerns about damaging the device. This anxiety is particularly pronounced when they are first introduced to tablets. Furthermore, Chiu and Liu (2017) noted that emotional attitudes and psychosocial difficulties are major barriers to using ICT in learning, often outweighing skill deficits. Similarly, Hage et al. (2020) has the same viewpoints, indicating that older adults not only face cognitive stress but also experience negative emotions and value conflicts when adapting to online media, which are critical factors in their ability to cope with technology-related challenges.

Structural factors

Environmental, resource, and social influence challenges further constrained the learning experiences of older adults. “Environmental context and resource challenges” were highlighted in nine studies. For instance, Chen et al. (2021) noted that many older adults lack access to tablets, primarily due to cost and insufficient internet access. Also, Baker et al. (2017) found that older adults in rural and disadvantaged areas face difficulties connecting to cellular networks and maintaining prepaid data plans, with financial constraints posing an additional significant challenge.

Lastly, the term “social influences” was examined in eight studies. Seo et al. (2019) observed that computer courses offered by technology-focused non-profit organizations often lack focus and relevance, prioritizing hardware over software use and other topics of interest to older adults. Then, more critically, beyond the course content issue, instructors frequently lack empathy and patience. Chen et al. (2021) further highlighted that older adults often miss opportunities to acquire digital skills due to insufficient support from instructors or family members, some of whom are unable to assist due to other commitments. These factors collectively contribute to the challenges older adults face in improving their DL.

The findings of this review align with recent studies in the field of DL in lifelong learning for older adults. Schreurs et al. (2017) suggests that older adults lag behind in their use of digital technologies, which limits social and institutional support systems and hinders their ability to gain the necessary technological experience and adaptability. Additionally, Nedeljko et al. (2022) demonstrated that there is a significant digital divide among older people in terms of inequality in DL, competence, and ICT use. Furthermore, Shi et al. (2023) indicated that the deterioration of vision, hearing, physical mobility, and memory in older people as they age makes it challenging for them to access information electronically. Multiple factors, such as older people’s socio-economic status, internet access, and poor ICT adaptation, greatly impact their development of DL, driving social exclusion and limiting participation for this particular age group. Therefore, to help bridge the gray digital divide and promote digital inclusion, it is essential for educators to approach the issue holistically, considering the aforementioned constraints that older people face daily.

Effective approaches to foster older adults’ DL

While lecture-based instruction is the most commonly employed method, it may not effectively foster DL among older adults. Many participants reported feeling overwhelmed or disengaged during these sessions. A teacher-centered approach guides students in a predetermined manner (Abarghouie et al., 2020) and efficiently delivers factual information to a large-group audience (Alaagib et al., 2019). However, its effectiveness can be compromised by the diverse physical and cognitive abilities, emotional states, and varying knowledge levels of older learners. A significant limitation is its failure to accommodate individual differences in digital knowledge and skills (Mayhorn et al., 2004).

Older adults, often hindered by barriers to accessing the digital world, may require more detailed instructions to fully understand and master desired digital skills. Mayhorn et al. (2004) found that older individuals tend to forget how to perform tasks learned in a digital training course when practicing on their own at home. This suggests that reliance on quick information processing and memory can present learning challenges for those experiencing physical and cognitive declines. To enhance DL training for older adults, prior research

recommends optimizing courses through smaller class sizes, grouping participants by skill level, providing step-by-step instruction, and maintaining a teaching pace suitable for older learners. Four essential methods were suggested: intergenerational learning, private tutoring, online interactive support, and game-based learning, demonstrating that age-appropriate, collaborative, and tailored training can empower older learners and foster a positive attitude toward using digital technologies.

Intergenerational learning can significantly enhance DL for older adults, facilitating knowledge exchange between generations in both formal and informal settings (Ropes, 2011). Wu and Chiou (2023) noted that intergenerational learning not only fosters a sense of purpose among older adults but also allows them to share their life experiences, reducing feelings of loneliness and social isolation. Zaid et al. (2022) demonstrated that intergenerational learning effectively enhances DL, encouraging collaboration with younger generations and improving cognitive skills and technological proficiency. Children often possess a better understanding of digital technologies, exhibit patience in teaching, and respect older adults, resulting in increased self-confidence and positive attitudes toward technologies among older learners.

Personal tutoring plays a crucial role by enabling tutors to closely monitor each student's learning progress. Earwaker (1992) describes this role as one of careful oversight. Tutors must demonstrate tasks step-by-step and provide guidance as needed (Por & Barriball, 2008). Seaton et al. (2023) found that older adults often lack one-on-one support in learning DL, while Taieb-Maimon and Vaisman-Fairstein (2022) noted that older individuals typically require more time and assistance to acquire basic digital skills. Personal tutoring can create tailored learning programs that address individual needs and consider the psychological and physical differences of older learners.

Recent advancements in online interactive support have enhanced interactivity in DL in lifelong learning. These tools deliver instruction through online courses or platforms, utilizing text, videos, interactive exercises, and quizzes (Moore & Hancock, 2022). Taieb-Maimon and Vaisman-Fairstein (2022) found that such online tools enable older adults to learn at their own pace, enabling them to review materials and practice anytime. However, Chen et al. (2021) reported that older adults often feel anxious about using tablets, fearing that they might damage the devices. Many older learners frequently forget how to complete tasks when practicing at home (Taieb-Maimon & Vaisman-Fairstein, 2022). Consequently, while software tools facilitate home learning, their lack of confidence and memory lapse can cause anxiety and stress, leading to frustration and reduced effectiveness without teacher support.

Game-based learning employs educational games that adapt to the learner's skill level, making the experience both motivating and enjoyable (Barzilai & Blau, 2014). Blažič and Blažič (2020) highlighted that game-based learning offers significant advantages in attractiveness and ease of use compared to traditional methods. However, cognitive load—referring to the overwhelming amount of information presented—can impede learning due to limited short-term memory capacity (Wilhelm et al., 2013). High cognitive load can lead to confusion and extended learning times. Schrader and Bastiaens (2012) found that learning in playful environments could result in higher cognitive loads, affecting knowledge retention and transferability. While this study did not determine whether game-based learning increases cognitive load for older adults, it suggests that cognitive challenges may arise in applying games to DL in lifelong learning for this demographic.

Awareness of ageist stereotypes and biases should be central to program design to ensure initiatives reflect the complex realities older adults face. Several studies highlight the benefits of tailored DL programs. Seo et al. (2019) highlighted how an evidence-based

DL class enables low-income African-American older adults to overcome socio-economic barriers and acquire essential digital skills. Boontasri and Temdee (2022) developed a personalized recommendation model that enhanced social network skills among seniors, emphasizing the importance of customization in educational programs. Additionally, Seaton et al. (2023) demonstrated that the Gluu Essentials Digital Skills Training made digital skills more accessible for the middle-aged and the senior participants, resulting in significant skill retention post-intervention.

Benefits of learning DL for older adults: Attitudinal, cognitive, social and health-related dimensions

RQ5 presents compelling evidence that well-designed learning strategies can effectively improve older learners' attitudes, confidence, competence, technology use habits and social engagement, and even their mental and cognitive health. This section synthesizes valuable insights from section "[Learning outcomes of DL learning programs for older learners \(RQ5\)](#)" into key benefits of digital learning for older adults. Most studies indicate that older learners have successfully developed digital knowledge over time through structured training programs which encompass cognitive, social, attitudinal/emotional, and health-related dimensions.

Firstly, from the cognitive dimension, studies reported that older adults were able to develop DL knowledge and skills at the end of their training programs. For instance, Gamliel and Gabay (2014) found that intergenerational learning facilitated knowledge exchange and empowerment among young people and older adults, which can significantly develop their DL in using different digital tools. Except for the intergenerational method, Chen et al. (2021) reported another method in using technological devices could also work, which found tablet usage for DL training was well-received by older adults with cognitive impairments and helped them overcome technical challenges. Similar to Chen et al.'s study, Wang et al. (2024) showed that the CATCH-ON connected tablet intervention effectively enhanced older adults' DL. Not the hardware, but the software, such as instant communication apps, also matters. Additionally, Vaportzis et al. (2017) found that older adults can develop their problem-solving skills and information processing abilities in using digital tools.

Secondly, DL training does enhance older adults' social engagement by fostering stronger connections with their family or friends, and it further encourages them to participate in online community activities. These training sessions shared the characteristics of learning how to use digital platforms like social media and instant messaging apps. Through these platforms, Kim et al. (2022) found that older adults could expand their social networks with all age groups, promoting mutual support and communication. Additionally, intergenerational learning experiences facilitated community building and communication between older generations and younger generations (Aldaco-Arias & Silas-Casillas, 2021). This increased interaction not only alleviated feelings of isolation—especially for those who live alone—but also enhanced social well-being and overall quality of life. Online community engagement empowers older adults by keeping them socially active, facilitating them to share their accumulated wisdom, and participate in meaningful activities and conversations that foster the exchange of ideas and stories.

Thirdly, a notable attitudinal/emotional shift can be found in the studies. Many older adults initially experienced anxiety and fear related to technology because of a lack of digital knowledge and confidence (Taieb-Maimon & Vaisman-Fairstein, 2022). However,

participation in well-structured training programs facilitated a transformation in their attitudes. For instance, Jurczyk-Romanowska et al. (2014) found that location-based games made DL learning immersive, significantly enhancing participants' enjoyment and engagement with the material. Similarly, Ferreira et al. (2016) noted that personalized support at educational telecenters helped older learners overcome initial resistance and develop a positive outlook toward digital inclusion. Baric et al. (2019) and Barbosa Neves et al. (2019) emphasized that viewing DL as a co-constructive learning process can help reduce feelings of isolation and loneliness among older adults. Also, Arthanat (2021) demonstrated that ICT adoption programs aimed at aging-in-place increased older adults' acceptance of technology and boosted their confidence in using it. Supporting these findings, Stanley et al. (2019) and Tsai et al. (2019) confirmed that age-friendly applications and structured learning environments offer more opportunities for older learners to further explore digital technologies and the digital world.

Lastly, cognitive and mental health improvement is also noteworthy. Cognitive health refers to the ability to think, learn, and remember clearly, which is crucial for effectively carrying out everyday activities and is a key aspect of overall brain health (Hertzog et al., 2008). Engaging in digital activities may provide cognitive benefits; for instance, Park et al. (2014) found that older adults who learned digital photography experienced greater memory improvement than those who engaged in less cognitively demanding activities. Chan et al. (2016) and Myhre et al. (2017) confirmed that interacting with digital technologies may enhance episodic memory, attention, and overall mental sharpness. Mental health is a state of well-being that enables individuals to cope with stresses, realize their potential, learn effectively, work productively, and contribute to their community. It holds intrinsic and instrumental value, and it is essential to the overall well-being of a person (WHO, 2022). Regarding mental health, Aldaco-Arias and Silas-Casillas (2021) enhanced the social inclusion of older adults by fostering a sense of community beyond the learning platform and encouraging intergenerational mutual learning. Similarly, Elbaz et al. (2024) found that a virtual DL program helped older adults maintain social connections and access to mental health support during the COVID-19 pandemic.

Conclusion, limitations, and recommendations

In conclusion, this review highlights a significant increase in studies focused on developing DL learning programs for older adults, with nearly three times as many published over the past decade. This trend parallels the rise in global digital technology usage, emphasizing the urgent need for older adults to adapt to these advancements. The disparity between high-income and low-middle-income countries reveals a greater level of technological adoption among older adults in wealthier nations, where more resources and support are available. Additionally, this review positions the DigComp 2.2 Framework within the context of older adults, reframing the 5 essential skills—problem solving, safety, digital content creation, information literacy, and communication—as tools to empower them alongside younger generations. Key challenges faced by older adults, including personal skill gaps and social influences, must be addressed for effective digital engagement.

The review further explores various teaching formats for DL in lifelong learning, emphasizing the importance of social, collaborative, and intergenerational support. Well-designed learning strategies should consider not only the attitudes and needs of older adults, but also their specific challenges at both the individual level and the structural level.

To address these challenges, this review found that older adults learn better in small classes with peers of similar skill levels, through step-by-step instructions, and at a measured pace. Effective methods for older adults include intergenerational learning, private tutoring, online interactive support, and game-based learning. By removing ageist stereotypes from program design, these methods can be optimized to address the specific challenges older adults face. Ultimately, structured DL training programs can significantly enhance DL and improve the quality of life for older adults in the digital age.

Limitations

This review article identifies key aspects of DL in lifelong learning for older adults; however, several areas necessitate further investigation to improve the effectiveness of these programs. Notably, the review lacks a comprehensive analysis of which specific approaches or interventions are more effective in facilitating the acquisition of various DL skills. Future research should focus on evaluating and comparing the impacts of different interventions on skill development. Additionally, the exclusion of non-English articles may limit the scope of our findings, as literature in other languages could offer valuable insights that might alter our conclusions. Furthermore, emerging phenomena, such as the influence of AI on older adults' awareness of scams, have yet to be addressed in the existing literature, leaving important gaps in evidence. Despite this limitation, our findings underscore the importance of functional dimensions and critical thinking in DL in lifelong learning, highlighting both the depth of our full-text review and the need for more comprehensive research in this area.

Recommendations for practitioners

By drawing on the findings in this review, several recommendations can be made to enhance the effectiveness of digital learning programs for older adults. First, implementing and evaluating programs on a long-term basis will ensure older learners have sufficient time to develop all-rounded digital skills with support to their physical, cognitive, social, and emotional needs during the lessons. Second, personalizing learning experiences for older adults of similar skill levels in small group settings will not only facilitate communication and collaborations, but also help enhance their engagement with other students and the content itself, maximizing the learning outcomes. Third, designing programs that address any practical barrier older adults may face will facilitate their regular use; an example of this is to combine digital cognitive training with physical exercise to promote hand-eye motor coordination when operating technological devices. By implementing these recommendations, learning programs can become more effective, inclusive, and sustainable for older adults.

Recommendations for future research

Future research should focus on several key areas to enhance the rigour of the research design. First, future research should prioritize geographic diversity by fostering targeted collaborations and including underrepresented contexts. Funding should emphasize representation to encourage studies in underexplored regions. A globally inclusive approach can enhance understanding and lead to more tailored DL programs for older adults worldwide.

Second, customizing and personalizing learning materials to address specific needs and preferences of different demographics, such as older adults in East Asia, can optimize training outcomes for specific locations and ensure their global replicability. Third, research should examine how older adults integrate digital technologies into their daily routines in post-research contexts, identifying any barriers in regular use so as to examine any potential implementation issue and scalability of these programs. Finally, assessing the potential of emerging technologies like virtual reality and generative AI in making all-rounded enhancements on older adults' health and well-being (i.e. their cognitive and mental functions, social interactions, and emotional well-being). By addressing these areas, future research can significantly enhance the quality of life for older learners, paving the way for the construction of a healthy aging society.

Appendix: Reviewed studies on digital literacy related to older adults (N = 46)

	Study	Intervention country	Research method	Key competencies	Challenges	Strategies	Learning outcomes
1	Gamliel and Gabay (2014)	Israel	Mixed	Foster effective communication and collaboration between older adults and children; Effective use of mobile/computer devices; Effective use of mobile applications; Use communication tools	Physical and cognitive limitations	Intergenerational learning	Confident and self-efficacy; Social engagement; Information Literacy
2	Jurczyk-Romanowska et al. (2014)	Poland	Quantitative	Entertainment themselves; Facilitate meaningful interactions and communication between older adults and others	Technical knowledge and skills; Physical and cognitive limitations	Game based learning	Attitude; Social engagement; Information Literacy

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
3	Chan et al. (2016)	USA	Quantita- tive	Effective use of mobile/com- puter devices; Effective use of mobile applications; Learn to use online tutorials and support resources; Apply digital skills to solve everyday problems; Troubleshoot basic computer related issues e.g., tablets usage and net- work problems	Physical and cognitive limitations	Lecture- Based Instruc- tion	Digital literacy skills; Social engage- ment/ Cognitive health; Infor- mation Literacy
4	Ferreira et al. (2016)	Brazil	Qualitative	Effective use of mobile/com- puter devices; Effective use of mobile applications; Capture and edit digital photos, audio, and video; Use com- munication tools; Explore digital creativ- ity tools and applications; Learn to use online tutorials and support resources; Use communica- tion tools like email; Use communica- tion tools	Technical knowledge and skills; social influences	Lecture- Based Instruc- tion	Overcoming Fear of learning technol- ogy; Digi- tal literacy skills; Attitude; Social engage- ment; Infor- mation Literacy

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
5	Taha et al. (2016)	USA	Quantita- tive	Organize and manage digital files and documents Interpret data visualizations and infograph- ics; Create and edit basic text documents, spreadsheets, and presenta- tions	Technical knowledge and skills	Software Tools	Digital literacy skills; Infor- mation Literacy
6	Baker et al. (2017)	Australia	Mixed	Effective use of mobile applications; Use commu- nication tools; Facilitate meaningful interactions and communi- cation between older adults and others	Technical knowledge and skills; Environ- mental context and resources; Environ- mental context and resources	Personal Tutoring	Confident and self- efficacy; Mental health; Social engage- ment
7	Delello and McWhorter (2017)	USA	Mixed	Effective use of mobile/com- puter devices; Effective use of mobile applications	Physical and cognitive limitations; Technical knowledge and skills	Lecture- Based Instruc- tion	Promotion of family relations; Digital literacy skills; Attitude; Social engage- ment; Cognitive health; Infor- mation Literacy

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
8	Gatti et al. (2017)	Italy	Mixed	Effective use of mobile/com- puter devices; Effective use of mobile applications; Facilitate meaningful interactions and communi- cation between older adults and others	Emotional factors; Physical and cogni- tive limita- tions;	Lecture- Based Instruc- tion	Confident and self- efficacy; Attitude; Overcom- ing Fear of learning technol- ogy; Digi- tal literacy skills; Motiva- tion; Social engage- ment; Informa- tion Literacy
9	Miwa et al. (2017)	Japan	Quantita- tive	Effective use of mobile/com- puter devices; Facilitate meaningful interactions and communi- cation between older adults and others	Environmen- tal context and resources	Lecture- Based Instruc- tion	Digital literacy skills; Confident and self- efficacy; Attitude; Social engage- ment
10	Myhre et al. (2017)	USA	Quantita- tive	Effective use of mobile/com- puter devices; Effective use of mobile applications	Physical and cognitive limitations	Lecture- Based Instruc- tion	Digital literacy skills; Social engage- ment; Cognitive health
11	Vaportzis et al. (2017)	UK	Quantita- tive	Effective use of mobile/com- puter devices; Effective use of mobile applications	Physical and cognitive limitations	Lecture- Based Instruc- tion	Digital literacy skills; Cognitive health; Informa- tion Literacy

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
12	Leone et al. (2018)	Canada	Mixed	Effective use of mobile/computer devices; Effective use of mobile applications; Facilitate meaningful interactions and communication between older adults and others	Physical and cognitive limitations; Social influences	Personal Tutoring	Motivation; Confident and self-efficacy; Mental health; Social engagement
13	Baric et al. (2019)	Sweden	Qualitative	Effective use of mobile/computer device; Effective use of mobile applications	Physical and cognitive limitations; Technical knowledge and skills	Software Tools	/
14	Barbosa Neves et al. (2019)	Canada	Mixed	Effective use of mobile/computer devices; Effective use of mobile applications; Use communication tools; Facilitate meaningful interactions and communication between older adults and others; Effectively search for and identify reliable online sources of information; Apply digital skills to solve everyday problems	Technical knowledge and skills; Emotional factors; Social influences	Lecture-Based Instruction	Digital literacy skills; Motivation; Attitude; Social engagement

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
15	Pachis and Zonneveld (2019)	Canada	Mixed	Effective use of mobile/computer devices; Effective use of mobile applications; Use communication tools; Facilitate meaningful interactions and communication between older adults and others; Effectively search for and identify reliable online sources of information; Apply digital skills to solve everyday problems	Physical and cognitive limitations	Software Tools	Digital literacy skills; Information Literacy
16	Seo et al. (2019)	USA	Mixed	Effective use of mobile/computer devices; Effective use of mobile applications; Troubleshoot basic computer related issues e.g., tablets usage and network problems; Facilitate meaningful interactions and communication between older adults and others; Understand the importance of personal data privacy and protection	Technical knowledge and skills; Environmental context and resources; Emotional factors; Social influences	Lecture-Based Instruction	Ability to recognise fake news and scams; Digital literacy skills; Motivation; Confident and self-efficacy; Social engagement; Information Literacy

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
17	Stanley et al. (2019)	USA	Mixed	Effective use of mobile/computer devices; Facilitate meaningful interactions and communication between older adults and others	Technical knowledge and skills	intergen- erational learning	Digital literacy skills; Motivation; Comfort with technology; Promotion of family and social relations; Attitude; Social engagement
18	Tsai et al. (2019)	USA	Quantita- tive	Effective use of mobile applications; Facilitate meaningful interactions and communication between older adults and others; Effectively search for and identify reliable online sources of information; Use communication tools	Technical knowledge and skills	Lecture- Based Instruc- tion	Digital literacy skills; Attitude; Social engagement; Information Literacy
19	Blažič and Blažič (2020)	Mixed	Mixed	Effective use of mobile/computer devices; Effective use of mobile applications; Entertainment themselves; Troubleshoot basic computer related issues; Facilitate meaningful interactions and communication between older adults and others	Physical and cognitive limitations; Emotional factors	Game based learning	Digital literacy skills; Attitude; Social engagement

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
20	Hage et al. (2020)	Nether- lands	Qualitative	Effective use of mobile/com- puter devices; Effective use of mobile applications	Emotional factors	Lecture- Based Instruc- tion	Confident and self- efficacy/ Overcom- ing Fear of learning technol- ogy/Digi- tal literacy skills; Infor- mation Literacy
21	Neil-Sztram- koet al. (2020)	Canada	Quantita- tive	Effective use of mobile/com- puter devices; Effective use of mobile applications	Emotional factors; Technical knowledge and skills	Lecture- Based Instruc- tion	Digital literacy skills; Promotion of family and social relations; Attitude; Social engage- ment; Infor- mation Literacy
22	Ten Brinke et al. (2020)	Canada	Quantita- tive	Effective use of mobile/com- puter devices; Effective use of mobile applications; Entertainment themselves	Technical knowledge and skills; Physical and cogni- tive limita- tions	Game based learning	Digital literacy skills; Cognitive health
23	Aldaco-Arias and Silas- Casillas (2021)	Mexico	Qualitative	Effective use of mobile/com- puter devices; Foster effective communi- cation and collaboration between older adults and children	Technical knowledge and skills	Intergen- erational learning	Digital literacy skills; Social engage- ment

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
24	Arthanat (2021)	USA	Mixed	Learn to use online tutorials and support resources; Use commu- nication tools; Effective use of mobile/com- puter devices; Effective use of mobile applications; Facilitate meaningful interactions and communi- cation between older adults and others	Emotional factors; Technical knowledge and skills; Physical and cogni- tive limita- tions	Personal Tutoring	Confident and self- efficacy; Digital literacy skills; Social engage- ment; Informa- tion Literacy
25	Castro Rojas (2021)	Costa Rica	Qualitative	Effective use of mobile/com- puter devices; Effective use of mobile applications; Use commu- nication tools; Entertainment themselves	Technical knowledge and skills; Physi- cal and cognitive limitations; Emotional factors; Environ- mental context and resources; Social influences	Lecture- Based Instruc- tion	Digital literacy skills; Social engage- ment; Cognitive health
26	Chen et al. (2021)	Hong Kong	Mixed	Effective use of mobile/com- puter devices	Technical knowledge and skills; Emotional factors; Environ- mental context and resources	Personal Tutoring	Confident and self- efficacy; Digital literacy skills; Overcom- ing Fear of learning technology and anxi- ety; Social engage- ment/ Cognitive health/ Mental health

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
27	Fields et al. (2021)	USA	Mixed	Effective use of mobile/computer devices; Effective use of mobile applications; Secure devices with strong passwords and update software regularly; Identify and avoid online scams, phishing attempts, and malware; Effectively search for and identify reliable online sources of information; Use communication tools; Apply digital skills to solve everyday problems; Troubleshoot basic computer related issues e.g., tablets usage and network problems; Learn to use online tutorials and support resources; Adapt to digital tools in everyday lives	Technical knowledge and skills	Personal Tutoring	Digital literacy skills; Confident and self-efficacy; Mental health

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
28	Holguin-Alvarez et al. (2021)	Peru	Quantita- tive	Effective use of mobile/computer devices; Use communication tools; Troubleshoot basic computer related issues e.g., tablets usage and network problems; Understand the importance of personal data privacy and protection; Foster effective communication and collaboration between older adults and children	/	Intergen- erational learning	Digital literacy skills; Social engagement; Information Literacy
29	Leek (2021)	Sweden	Mixed	Effective use of mobile/computer devices; Effective use of mobile applications; Learn to use online tutorials and support resources; Facilitate meaningful interactions and communication between older adults and others; Apply digital skills to solve everyday problems (e.g., learning how to manipulate with various mobile app	Technical knowledge and skills	Intergen- erational learning	Digital literacy skills; Social engagement; Information Literacy

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
30	Rohner et al. (2021)	Austria	Qualitative	Effective use of mobile/com- puter devices; Effective use of mobile applications	Technical knowledge and skills; Physi- cal and cognitive limitations; Emotional factors; Environ- mental context and resources	Intergen- erational learning	Digital literacy skills; Attitude; Social engage- ment
31	Boontasri and Temdee (2022)	Thailand	Quantita- tive	Effective use of mobile/com- puter devices; Effective use of mobile applications	Physical and cognitive limitations	Software Tools	Digital literacy skills
32	Choi and Park (2022)	South Korea	Quantita- tive	Entertainment themselves	Technical knowledge and skills	Game based learning	Digital literacy skills
33	Kim et al. (2022)	USA	Quantita- tive	Effective use of mobile/com- puter devices; Use communi- cation tools	Technical knowledge and skills; Emotional factors; Environ- mental context and resources	Lecture- Based Instruc- tion	Promotion of family and social relations; Digital literacy skills; Social engage- ment
34	Taieb-Mai- mon and Vaisman- Fairstein (2022)	Jerusa- lem	Quantita- tive	Effective use of mobile/com- puter devices; Effective use of mobile applications	Technical knowledge and skills; Physi- cal and cognitive limitations; Emotional factors	dialogic learning approach	Digital literacy skills; Confident and self- efficacy; Attitude; Mental health; Infor- mation Literacy
35	Korpela et al. (2023)	Finland	Qualitative	Effective use of mobile appli- cations; Apply digital skills to solve everyday problems	Physical and cognitive limitations	Peer learn- ing	Digital literacy skills; Mental health

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
36	Liang et al. (2023)	China	Quantita- tive	Effective use of mobile appli- cations	/	Lecture- Based Instruc- tion	Promotion of family and social relation; Mental health; Infor- mation Literacy
37	Seaton et al. (2023)	Canada	Mixed	Effective use of mobile/com- puter devices; Effective use of mobile applications; Practice safe online behaviours; Troubleshoot basic computer related issues; Adapt to digital tools in everyday lives; Effectively search for and identify reliable online sources of information	Environmen- tal context and resources; Technical knowledge and skills; Social influences	Software Tools	Digital literacy skills; Infor- mation Literacy
38	Tsugawa (2023)	Mixed	Qualitative	Effective use of mobile/com- puter devices; Effective use of mobile applications; Use communi- cation tools	Emotional factors; Technical knowledge and skills	Lecture- Based Instruc- tion	Motivation; Digital literacy skills; Social engage- ment; Infor- mation Literacy

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
39	Zhang et al. (2023)	China	Qualitative	Effective use of mobile/computer devices; Effective use of mobile applications; Practice safe online behaviors; Use communication tools; Facilitate meaningful interactions and communication between older adults and others; Apply digital skills to solve everyday problems; Identify and avoid online scams, phishing attempts, and malware; Adapt to digital tools in everyday lives; Apply digital skills to solve everyday problems	Environmen- tal context and resource; Emotional factors; Technical knowledge and skills	Personal Tutoring	Digital literacy skills; Promotion of family and social relations; Social engage- ment; Infor- mation Literacy
40	Elbaz et al. (2024)	Canada	Quantita- tive	Effective use of mobile/computer devices; Effective use of mobile applications; Use communication tools	Social influ- ences	Lecture- Based Instruc- tion	Digital literacy skills/ Comfort with tech- nology; Infor- mation Literacy

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
41	Ferguson et al. (2024)	USA	Mixed	Effective use of mobile/computer devices; Identify and avoid online scams, phishing attempts, and malware; Adapt to digital tools in everyday lives; Use communication tools; Practice safe online behaviors	Technical knowledge and skills; Physical and cognitive limitations	Personal Tutoring	Information Literacy
42	Leedahl et al. (2024)	USA	Mixed	Effective use of mobile/computer devices	Technical knowledge and skills; Physical and cognitive limitations; Emotional factors; Environmental context and resources	Intergenerational learning	Digital literacy skills; Promotion of family and social relations; Comfort with technology; Social engagement; Mental health; Information Literacy
43	Saravanan et al. (2024)	USA	Qualitative	Effective use of mobile/computer devices; Troubleshoot basic computer related issues; Effective use of mobile applications	Technical knowledge and skills; Emotional factors	Lecture-Based Instruction	Confident and self-efficacy; Attitude; Digital literacy skills; Social engagement; Mental health; Information Literacy

	Study	Inter- vention country	Research method	Key competen- cies	Challenges	Strategies	Learning outcomes
44	Tellado et al. (2024)	Spain	Qualitative	Effective use of mobile/computer devices; Effective use of mobile applications; Facilitate meaningful interactions and communication between older adults and others; Capture and edit digital photos, audio, and video	Physical and cognitive limitations	dialogic learning approach	Digital literacy skills; Promotion of family and social relations; Social engagement; Mental health; Information Literacy
45	Wang et al. (2024)	USA	Quantita- tive	Effective use of mobile/computer devices; Effective use of mobile applications; Communicate with others via web-conferencing software; Learn to use online tutorials and support resources e.g., navigating the Internet and accessing online resource	Social influ- ences	Lecture- Based Instruc- tion	Digital literacy skills; Confident and self-efficacy; Social engagement; Mental health; Information Literacy
46	Moore and Hancock (2022)	USA	Quantita- tive	Identify and avoid online scams, phishing attempts, and malware	Emotional factors	Software Tools	digital literacy skills; Information Literacy

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Declarations

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